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Waste disposal and management practices in fruit and vegetable processing plants in Gujarat

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Abstract

The growing demand for processed fruits and vegetables has significantly increased waste generation in processing plants, raising environmental and operational concerns. This study investigates the current waste disposal and management practices employed in Processing of Fruits and Vegetables facilities across Gujarat, a state recognized for its vibrant agro-processing sector. The research aims to evaluate the efficiency, sustainability, and compliance of existing waste management systems with environmental regulations. Primary and secondary data were collected from selected processing units, interviews with plant managers, and government reports. The types of waste analyzed include peels, seeds, pulp residues, and wastewater. Findings reveal that while some large-scale facilities implement composting, biogas production, and wastewater treatment, many small- and medium-scale units still rely on conventional disposal methods that contribute to environmental degradation. Lack of awareness, limited access to technology, and cost constraints were identified as major barriers to adopting sustainable practices. The study underscores the need for policy interventions, capacity building, and investment in green technologies to improve waste management in the sector. It also highlights the potential for converting organic waste into value-added products, promoting a circular economy model. This research contributes to understanding the gaps and opportunities in waste management within the Processing of Fruits and Vegetables industry in Gujarat and offers practical recommendations for stakeholders to enhance sustainability and regulatory compliance.

Keywords Disposal practices, Sustainability, Organic waste, Circular economy, Agro-industries, Green technologies

1 Introduction

The management of waste generated from Fruit And Vegetable Processing Plants is a pressing concern, particularly in agriculturally rich states like Gujarat. These industries, while contributing significantly to the agrarian economy, also produce substantial amounts of biodegradable waste, including peels, pulp, seeds, and wastewater [1]. In Gujarat, which ranks among the top states in horticultural output, the proliferation of



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small-scale processing units without systematic waste disposal methods poses a serious environmental challenge [2, 3].

Studies have shown that in many regions of Gujarat, the waste generated from these plants is often either discarded indiscriminately or occasionally repurposed informally by local farmers as animal feed or organic compost [4]. However, these practices lack standardization and fail to address the broader environmental and health implications of improper waste handling, such as water contamination and soil degradation [5, 6].

the absence of integrated waste management strategies and inadequate infrastructure, such as effluent treatment plants and composting facilities, further exacerbates the issue. The state's vegetable and fruit waste recycling potential remains largely underutilized due to fragmented policies and limited adoption of sustainable technologies [7, 8]. The scenario calls for urgent attention to develop and implement organized, eco-friendly waste management practices within these processing units, aligned with circular economy principles [9].

In recent years, there has been growing interest in valorizing fruit and vegetable waste through composting, bio-enzymatic treatment, and biomethanation, which not only reduce environmental pollution but also create economic opportunities through the generation of biogas, organic fertilizers, and value-added products [10]. However, the uptake of these methods in Gujarat remains inconsistent and mostly limited to pilot initiatives.

This study aims to explore the existing waste disposal and management practices employed in Fruit And Vegetable Processing Plants in Gujarat, identify gaps and challenges, and recommend viable, sustainable solutions for improving the sector's environmental footprint [11]. The food processing industry is one of the fastest-growing sectors in India, contributing significantly to agricultural value addition and employment generation. Within this sector, Fruit And Vegetable Processing Plants play a vital role in reducing post-harvest losses and meeting the increasing demand for packaged food. However, these plants generate large quantities of organic and inorganic waste during washing, peeling, cutting, pulping, canning, and packaging stages [12]. Waste from such industries primarily consists of peels, seeds, pulp residues, spoiled raw material, wastewater, and packaging materials such as plastics and cans [13]. If not managed properly, this waste can result in environmental degradation, water pollution, greenhouse gas emissions, and adverse effects on community health [14].

In Gujarat, one of India's leading states in agro-processing, waste generation is particularly significant due to its high concentration of Processing of Fruits and Vegetables clusters in districts such as Anand, Sabarkantha, and Kheda. Recent estimates suggest that India generates nearly 50 million tonnes of municipal solid waste annually, of which only about 20–25% is processed or recycled, while the rest is either landfilled or dumped in open areas [15]. Within Gujarat, approximately 5–7% of this waste originates from agro-industrial activities, but less than 30% of the fruit and vegetable waste is reused through composting, animal feed, or energy recovery [13]. This indicates a substantial gap between waste generation and its sustainable utilization.

Given the rising emphasis on sustainable development, there is a growing need to examine how Fruit And Vegetable Processing Plants in Gujarat handle waste. Studying current practices, challenges, and policy implications can help design effective waste minimization and resource recovery strategies [16].

1.1 Research objectives

1. To assess the types and quantities of waste generated in Fruit And Vegetable Processing Plants in Gujarat.
2. To examine the existing waste disposal and management practices adopted by these processing plants.
3. To evaluate the effectiveness and sustainability of current waste management strategies.
4. To identify the challenges and barriers faced by processing plants in implementing efficient waste management practices.

1.2 Research questions

- What types and volumes of waste are commonly produced in Processing of Fruits and Vegetables units across Gujarat?
- What methods are currently employed by these plants for the disposal and management of processing waste?
- How effective and environmentally sustainable are the current waste management practices in these facilities?
- What are the major obstacles preventing the adoption of improved waste disposal and management systems in the industry?

2 Literature review

Waste management has been a subject of global concern due to its direct link to human health, environmental quality, and sustainable resource use. Giusti [14] highlighted that poor waste disposal practices contribute to soil, air, and water contamination, thereby increasing the risk of infectious diseases and long-term health hazards. Similarly, Hos-sain et al. [17] reviewed clinical solid waste management and emphasized that landfilling and open dumping remain dominant practices worldwide, though they are environmentally unsustainable.

In developing nations, challenges in waste management are often associated with weak infrastructure, lack of enforcement, and public awareness [18]. Studies in African cities show that even when formal disposal systems exist, residents and industries frequently practice open dumping due to limited accessibility and affordability of collection services [19]. Comparable trends are observed in Asian contexts, for instance, in Indonesia, Meidiana and Gamse [20] found that inadequate legal frameworks and enforcement mechanisms contributed to ineffective solid waste management [21].

India presents a unique case where municipal solid waste management is gradually shifting from disposal to resource recovery approaches such as composting, biogas production, and recycling [13]. However, the agro-processing industry continues to generate substantial biodegradable waste that is often underutilized. Aming-Hayudini et al. [22] observed that most municipal waste management systems prioritize household solid waste, with limited focus on agro-industrial residues, despite their high organic content and potential for compost or bioenergy production [23].

Several scholars have also highlighted the implications of poor waste management in industrial contexts. Adeniyi [24] noted that distance from disposal sites often leads to indiscriminate dumping by urban residents and small industries. Ampofo [25] further

showed that weak institutional monitoring in educational and industrial facilities often perpetuates unsustainable waste disposal practices [26]. During crises such as the COVID-19 pandemic, the challenges of waste management intensified, with increased volumes of disposable protective equipment and organic waste stressing existing systems [27].

Within the Indian food processing sector, waste utilization remains a challenge, although some progress has been reported. Studies suggest that fruit and vegetable waste can be effectively transformed into value-added products such as biogas, bioethanol, pectin, and animal feed [28]. Yet, the implementation of such practices in Gujarat remains limited, with most processing units relying on land disposal or informal sale of by-products to cattle farmers [8].

Overall, the literature indicates that while the environmental and health impacts of poor waste disposal are well-documented [14, 17], there is a lack of region-specific studies examining waste disposal practices within the agro-industrial sector in India, particularly in Gujarat. Addressing this gap is essential to understand the extent of waste utilization, barriers to sustainable management, and the role of policy and technology in improving practices [29].

2.1 Types and quantities of waste generated in fruit and vegetable processing plants in Gujarat

Processing of Fruits and Vegetables industries are significant contributors to food supply chains but are also major generators of both solid and liquid waste. Gujarat, a leading state in horticultural production, has seen a rapid expansion of processing units, which in turn has escalated the volume and complexity of waste generated. The waste largely comprises peels, seeds, stones, pulp residues, and wastewater, varying with the type of produce processed [2, 3]. These residues, while organic, pose serious environmental threats if not effectively managed.

Studies show that the quantity and type of waste vary significantly depending on the commodity. For example, tomato and onion processing units in Gujarat generate vast quantities of skin and pulp waste [7], while watermelon processing leads to large volumes of liquid waste due to its high water content [30]. A proximate analysis by Gupta et al. [4] on fruit and vegetable waste collected from the Kadi market in Gujarat revealed high moisture content in most samples, indicating a predominant generation of wet waste. These waste streams are often rich in bioactive compounds and nutrients, yet they remain underutilized [31].

The volume of waste is influenced not only by the produce type but also by the scale of operations and efficiency of processing technologies used. Processing steps like peeling, slicing, deseeding, and juicing inherently generate high volumes of by-products. Raj et al. [5] observed that up to 40% of raw input could be lost as waste in inefficient units. infrastructural limitations such as inadequate cold storage and logistics contribute to pre-processing waste, further adding to the total volume [32].

Despite the challenges, some efforts are underway to repurpose this waste for bioenergy, compost, and value-added products. However, a structured waste management system is still lacking. As Joshi [2] emphasizes, there is a need for integrated waste management strategies combining waste minimization, segregation at source, and recovery of bioactive compounds.

In conclusion, Gujarat's Processing of Fruits and Vegetables sector generates significant quantities of heterogeneous waste. Understanding the types and quantities of this waste is crucial for designing effective waste-to-value systems and sustainable processing models (Fig. 1).

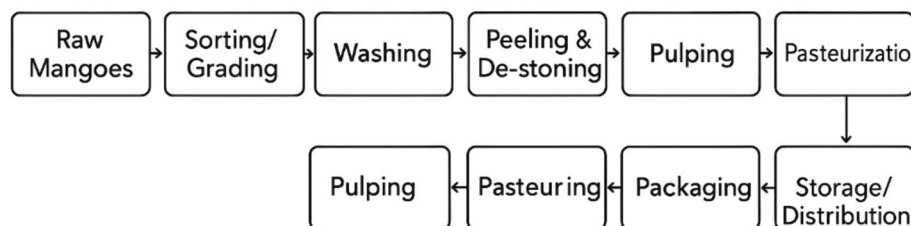
2.2 Challenges and barriers faced by processing plants in implementing efficient waste management practices

Efficient waste management remains a significant concern for processing plants across diverse industries, especially in developing and transitional economies. Numerous studies highlight systemic, infrastructural, financial, and regulatory barriers that hinder the adoption of sustainable practices. Alshdiefat et al. [33] emphasize that in countries like Jordan, processing facilities grapple with fragmented policies, insufficient stakeholder coordination, and a lack of skilled personnel, all of which obstruct the establishment of effective waste management systems. Similarly, Smol et al. [34] discuss how water and wastewater treatment plants encounter institutional inertia and lack of economic incentives when attempting to adopt circular economy principles.

In the broader construction and industrial sectors, resistance to change, high initial investment costs, and technological incompatibilities are recurring obstacles [35]. These are exacerbated by poor regulatory enforcement and minimal training among operational staff. The complexity of integrating waste management systems into existing production processes often discourages plant operators, particularly in regions with limited technical expertise [36, 37].

A further complication lies in the limited cultural and behavioral readiness to embrace sustainable waste practices. Digitemie et al. [38] assert that industrial operations in oil and gas sectors face socio-cultural barriers alongside logistical and economic constraints, necessitating a transformative shift in organizational mindset and values. This insight is echoed by Debrah et al. [39], who identify deep-rooted infrastructural gaps and public apathy as critical barriers in Sub-Saharan Africa.

Mango Pulp Processing and Waste Generation



Waste Generated at Each Step:

- Sorting/Grading → rejected/overripe fruits
- Washing → contaminated water, mud/dust
- Peeling & De-stoning → peels, stones, fibrous pulp residues
- Pulping → seed husk, fiber

Fig. 1 Step by step method of processing and waste generation

In rural or tourism-oriented regions, logistical hurdles such as inaccessible locations, inadequate waste collection systems, and seasonal variations in waste volume further challenge the implementation of efficient systems [40]. Additionally, the lack of proper data management, monitoring tools, and performance indicators undermines the operational effectiveness of waste processing initiatives [41].

Globally, the transition toward zero-waste manufacturing is impeded by technological limitations, fragmented policy frameworks, and limited inter-organizational collaboration [42]. These findings suggest that waste management in processing plants is not merely a technical challenge but a multifaceted issue requiring institutional support, financial investment, cultural transformation, and robust regulatory backing [43].

In conclusion, processing plants face a convergence of technical, institutional, behavioral, and economic barriers in implementing efficient waste management practices. Overcoming these challenges demands a holistic and integrative approach that combines policy reform, capacity building, stakeholder engagement, and technology adoption tailored to local and industrial contexts [44].

This study contributes to the literature by addressing an underexplored domain within waste management—the fruit and vegetable processing plants of Gujarat. While earlier research has largely concentrated on urban municipal waste or general agri-food byproducts, very few studies have examined the region-specific practices, limitations, and opportunities of processing units in Gujarat. By situating the analysis within this context, the study adds a geographical and sectoral novelty to the discourse.

Methodologically, the study adopts an integrated approach that combines assessment of current disposal practices with the evaluation of underutilized techniques such as solar drying, biochar production, and waste-to-value conversion. This dual lens provides a more comprehensive understanding of both conventional and emerging strategies, thereby extending the methodological scope of existing literature.

3 Methodology

3.1 Data collection

For this study, data was gathered from 100 unit managers working in Fruit And Vegetable Processing Plants located across various districts in Gujarat. The data collection process was carried out using a structured questionnaire developed to assess the existing waste disposal techniques, frequency of waste generation, awareness levels about sustainable waste management, and compliance with environmental standards.

Most studies focus on urban waste or generic food industries, but this study uniquely addresses small- and medium-scale fruit & vegetable plants in Gujarat.

To ensure the quality and relevance of the data, the questionnaire included both close-ended and open-ended questions. The close-ended items helped in obtaining quantifiable information, while the open-ended responses provided qualitative insights into innovative or plant-specific waste management strategies. Prior to full-scale deployment, a pilot test was conducted on a small group of managers to refine the language and structure of the instrument.

Primary data collection was done through direct visits to the units, telephonic interviews, and email responses, depending on the convenience and availability of the respondents. To maintain accuracy and consistency, all participants were assured of confidentiality, and their informed consent was obtained before participation.

3.1.1 Sampling method

The study employed a purposive sampling technique to select respondents. This non-probability sampling method was chosen due to the specific knowledge and experience required from the participants—namely, individuals directly involved in managing or supervising waste disposal operations within the processing units.

Given the diverse geographical distribution of Processing of Fruits and Vegetables units in Gujarat, care was taken to ensure representation from key production hubs, including districts such as Anand, Surat, Vadodara, and Rajkot. The selection criterion for inclusion in the sample was based on the manager's familiarity with plant-level waste management practices and their role in decision-making processes.

This targeted approach ensured that the sample included individuals who could provide reliable, detailed, and relevant information about waste management practices, making the findings more meaningful and actionable for policy recommendations.

3.1.2 Hypothesis

H1 There is significant impact of Waste Management Strategies On working efficiency of Processing of Fruits and Vegetables Plant Industry.

H2 There is significant difference between various methods employed by plants for the disposal and management of processing waste.

Dependent variables: Processing of fruits and vegetables plant industry

Independent variables: Waste management strategies, methods employed by plants for the disposal and management of processing waste

3.1.3 Statistical tools

SPSS Version 26 and Microsoft Excel is used to analyze the data and the test used in analysis are Multiple Regression, Friedman's Ranking Test, Descriptive analysis and One-way ANOVA test.

4 Data analysis

4.1 Waste management strategies on working efficiency of processing of fruits and vegetables plant industry

H1 There is significant impact of waste management strategies on working efficiency of processing of fruits and vegetables plant industry (Figs. 2, 3 and 4; Tables 1, 2, 3 and 4).

4.2 Various methods employed by plants for the disposal and management of processing waste

H2 There is significant difference between various methods employed by plants for the disposal and management of processing waste (Tables 5, 6 and 7).

5 Results and discussion

The study provides a detailed assessment of waste disposal and management practices in Fruit And Vegetable Processing Plants in Gujarat. The findings highlight both the progress made in structured waste management and the challenges that remain for sustainable implementation.

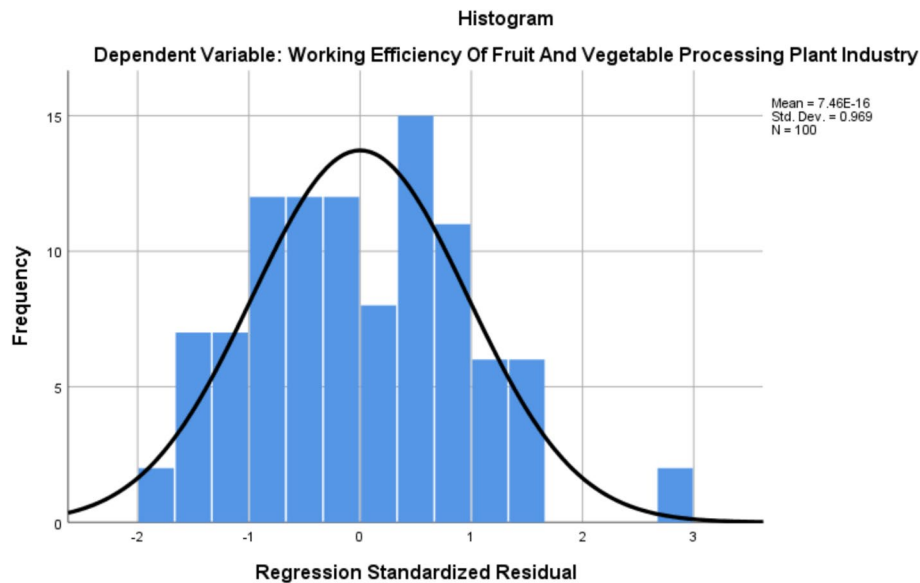


Fig. 2 Histogram on working efficiency of processing of fruits and vegetables plant industry

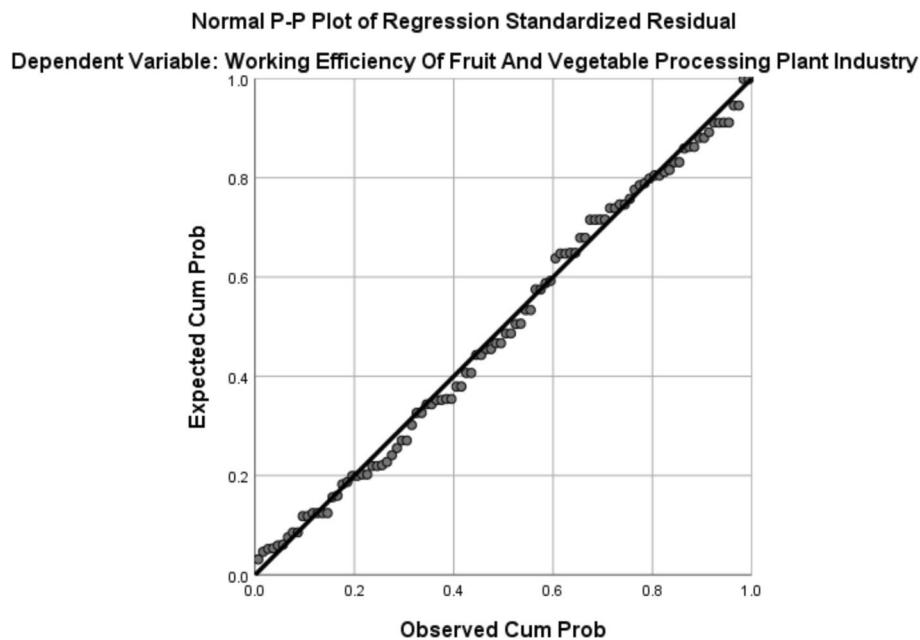


Fig. 3 Normal P-P plot on working efficiency of processing of fruits and vegetables plant industry

To strengthen interpretation, the numerical results are explained in terms of their practical significance. The statistical outcomes are compared with relevant literature, ensuring that the discussion not only interprets the results but also situates them within the wider body of research on industrial waste management [14, 17, 18].

5.1 Waste management strategies and working efficiency

The results clearly demonstrate a significant positive relationship between waste management practices and the operational efficiency of processing plants in Gujarat. Descriptive statistics indicated that employee training and structured waste practices

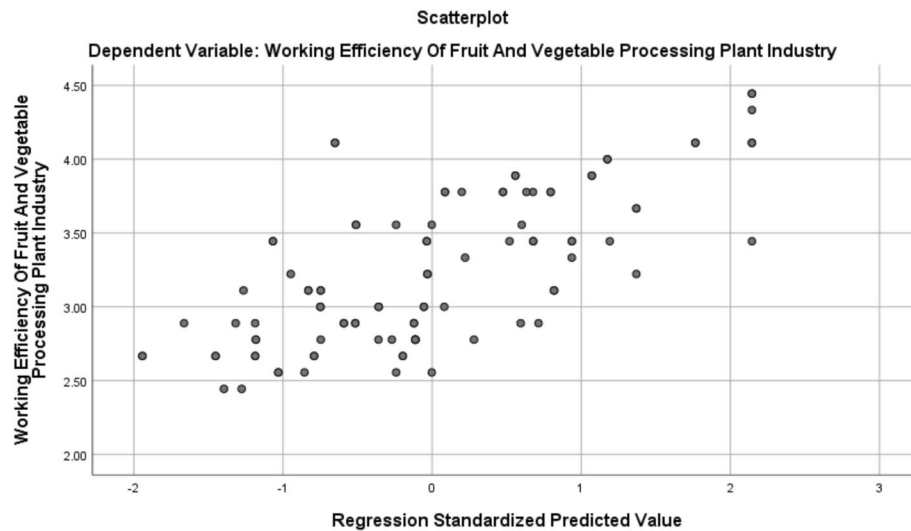


Fig. 4 Scatterplot on working efficiency of processing of fruits and vegetables plant industry

Table 1 Descriptive statistics

	Mean	Std. deviation	N
Working Efficiency Of Processing of Fruits and Vegetables Plant Industry	3.2467	0.51576	100
Efficient waste management practices contribute positively to the overall productivity of our processing plant	3.4500	0.84537	100
The implementation of structured waste disposal methods reduces downtime and operational disruptions	3.6900	1.27679	100
Our plant's working efficiency has improved since adopting eco-friendly waste management strategies	3.5000	1.18492	100
Poor waste handling adversely affects the workflow and hygiene standards in the processing unit	3.5900	1.15553	100
Training employees on waste segregation and disposal enhances the smooth functioning of daily operations	3.7600	1.15575	100
Investing in sustainable waste management technologies has led to measurable improvements in plant efficiency	2.4200	1.31180	100

Table 2 Model summary

Model	R	R square	Adjusted R square	Std. error of the estimate
1	0.718 ^a	0.516	0.485	0.37029

Dependent variable: working efficiency of processing of fruits and vegetables plant industry

^aPredictors: (Constant), Investing in sustainable waste management technologies has led to measurable improvements in plant efficiency., Training employees on waste segregation and disposal enhances the smooth functioning of daily operations., Efficient waste management practices contribute positively to the overall productivity of our processing plant., Poor waste handling adversely affects the workflow and hygiene standards in the processing unit., The implementation of structured waste disposal methods reduces downtime and operational disruptions., Our plant's working efficiency has improved since adopting eco-friendly waste management strategies

scored higher than other variables, suggesting that managers and workers recognize their importance in streamlining plant operations. This echoes the observations of Aming-Hayudini et al. [22], who noted that effective waste handling improves both productivity and workplace safety.

The regression analysis revealed a strong correlation ($R = 0.718$) and a coefficient of determination ($R^2 = 0.516$). In practical terms, this means that more than half of the variability in plant efficiency can be explained by waste management strategies. Comparable

Table 3 ANOVA

Model		Sum of squares	df	Mean square	F	Sig
1	Regression	13.584	6	2.264	16.511	0.000 ^a
	Residual	12.752	93	0.137		
	Total	26.335	99			

Dependent variable: working efficiency of processing of fruits and vegetables plant industry

^aPredictors: (Constant), Investing in sustainable waste management technologies has led to measurable improvements in plant efficiency., Training employees on waste segregation and disposal enhances the smooth functioning of daily operations., Efficient waste management practices contribute positively to the overall productivity of our processing plant., Poor waste handling adversely affects the workflow and hygiene standards in the processing unit., The implementation of structured waste disposal methods reduces downtime and operational disruptions., Our plant's working efficiency has improved since adopting eco-friendly waste management strategies

Table 4 Coefficients

Model		Unstandardized coefficients		Standardized coefficients	t	Sig
		B	Std. error			
1	(Constant)	1.731	0.190		9.125	0.000
	Efficient waste management practices contribute positively to the overall productivity of our processing plant	0.146	0.051	0.239	2.851	0.005
	The implementation of structured waste disposal methods reduces downtime and operational disruptions	−0.058	0.049	−0.144	−1.193	0.236
	Our plant's working efficiency has improved since adopting eco-friendly waste management strategies	0.044	0.055	0.102	0.809	0.420
	Poor waste handling adversely affects the workflow and hygiene standards in the processing unit	0.072	0.049	0.161	1.478	0.143
	Training employees on waste segregation and disposal enhances the smooth functioning of daily operations	0.088	0.038	0.198	2.323	0.022
	Investing in sustainable waste management technologies has led to measurable improvements in plant efficiency	0.199	0.032	0.506	6.222	0.000

Table 5 Descriptive statistics

	N	Mean	Std. deviation	Minimum	Maximum
Composting	100	3.1700	1.24766	1.00	5.00
Bio-methanation	100	4.1000	0.88192	1.00	5.00
Vermicomposting	100	3.5000	1.12367	1.00	5.00
Conversion into animal feed	100	4.1200	1.11265	1.00	6.00
Incineration	100	3.8900	0.95235	1.00	5.00
Landfilling	100	3.3700	0.82456	1.00	4.00
Wastewater treatment	100	3.3800	1.22911	1.00	5.00
Solar drying	100	1.2700	0.44620	1.00	2.00
Biochar production	100	2.4300	1.12146	1.00	4.00
Recycling of packaging waste	100	3.8000	1.18918	1.00	5.00

Table 6 Ranks

	Mean rank
Composting	5.33
Bio-methanation	7.30
Vermicomposting	5.82
Conversion into animal feed	7.29
Incineration	6.73
Landfilling	5.24
Wastewater treatment	5.60
Solar drying	1.64
Biochar production	3.54
Recycling of packaging waste	6.54

Table 7 Test statistics

N	100
Chi-square	365.274
df	9
Asymp. sig	0.000

findings were reported in the food processing sector globally, where systematic waste handling improved efficiency and reduced environmental risk [19, 24].

ANOVA results further confirmed the robustness of the model ($F = 16.511$, $p < 0.001$). The coefficient analysis highlighted three particularly influential variables:

- *Investment in sustainable waste management technologies* ($\beta = 0.506$, $p < 0.001$) – the strongest predictor, underscoring the role of modern technology in optimizing efficiency.
- *Training employees on waste segregation and disposal* ($\beta = 0.198$, $p = 0.022$) – aligning with Hantoko et al. [27], who emphasized capacity building as key to successful waste strategies.
- *Efficient waste management practices contributing to productivity* ($\beta = 0.239$, $p = 0.005$) – reinforcing the broader idea that waste minimization directly supports performance outcomes [20].

Overall, the hypothesis (H1) was supported. The evidence suggests that structured waste management, particularly investments in technology and training, plays a critical role in improving the working efficiency of Fruit And Vegetable Processing Plants in Gujarat.

5.2 Methods of waste disposal and management

The survey of 100 processing units revealed significant variation in the adoption of waste disposal methods. The most frequently employed strategies were conversion into animal feed (mean = 4.12) and bio-methanation (mean = 4.10). These sustainable practices indicate a growing awareness of circular economy principles in the industry. Similar patterns have been documented in European and Asian contexts, where bio-methanation and

feed conversion are considered economically viable and environmentally sound options [13, 14].

Other moderately adopted methods included incineration (mean = 3.89) and recycling of packaging waste (mean = 3.80). While these approaches contribute to reducing landfill dependency, they require stricter monitoring to avoid secondary environmental impacts, such as air emissions [45].

Conversely, solar drying (mean = 1.27) and biochar production (mean = 2.43) were least utilized, likely due to limited technical expertise, higher upfront costs, and lack of institutional support. Previous studies in developing regions confirm that such low-cost but less popular techniques remain underutilized without targeted policy interventions [25, 46].

The Friedman test confirmed significant differences among the methods ($\chi^2 = 365.274$, $p < 0.001$), validating Hypothesis (H2). Ranking results placed bio-methanation and animal feed conversion at the top, while solar drying ranked lowest. This hierarchy demonstrates the preference for methods with clear economic returns and proven efficiency, while innovative but less familiar techniques remain marginalized.

6 Conclusion

The findings provide valuable insights into the waste disposal and management practices of Fruit And Vegetable Processing Plants in Gujarat. Key takeaways include:

1. *Nature of waste* The waste is predominantly organic (peels, pulp, seeds, and biodegradable residues), generated in large volumes, particularly during peak seasons.
2. *Current practices* Larger plants show better adoption of structured systems like composting and bio-methanation, whereas smaller units still rely on basic methods such as open dumping and burning, which create environmental and health concerns [18, 47].
3. *Effectiveness of practices* Plants using structured waste management reported better environmental outcomes, but adoption remains uneven due to financial, technical, and logistical barriers.
4. *Challenges* High costs of technologies, lack of skilled staff, inadequate awareness of government schemes, and space constraints emerged as the main obstacles. These findings are consistent with broader studies on waste management challenges in developing contexts [15, 17].

In conclusion, while there is increasing recognition of the importance of sustainable waste management in Gujarat's Processing of Fruits and Vegetables sector, the transition is uneven. Policy interventions, financial incentives, and training programs are essential to encourage wider adoption of eco-friendly practices such as biochar production and solar drying. With coordinated action, the industry can move towards a more circular economy model, aligning operational efficiency with environmental sustainability.

7 Recommendations

Improving waste disposal and management in Gujarat's Processing of Fruits and Vegetables sector requires a multi-layered approach that combines policy support, technology, and stakeholder participation.

A priority step is to strengthen waste segregation at the source. Processing plants should be encouraged to classify waste into biodegradable, recyclable, and non-recyclable streams at the point of generation. This practice enhances the efficiency of subsequent treatment processes and maximizes the potential for recycling and reuse. Awareness campaigns and guidelines tailored to food processing units can ensure consistent adoption.

Given that most waste generated in these facilities is organic, adoption of eco-friendly technologies such as composting, vermicomposting, and bio-methanation is highly recommended. Converting organic waste into compost or biogas not only minimizes environmental hazards but also yields value-added products. The role of government is crucial here—by offering subsidies, tax incentives, or technical assistance under existing schemes like the *Swachh Bharat Mission* or *Zero Effect Zero Defect (ZED)* certification, the transition to sustainable systems can be made more feasible, especially for small and medium enterprises (SMEs) [48].

Capacity building and training programs are equally important. Many units lack adequately trained personnel for handling, monitoring, and managing waste in line with best practices. Regular workshops, on-site demonstrations, and certification courses can bridge this gap. Case studies highlighting success stories from well-performing plants may further motivate others to adopt improved methods [30].

Public-private partnerships and NGO involvement offer another promising avenue. Collaborating with specialized waste management firms or non-governmental organizations can provide technical expertise, logistical support, and awareness-building at the community level. Such partnerships also help extend waste collection and processing services to regions with limited public infrastructure[49].

Embedding the principles of a circular economy within operational practices can shift the perception of waste from a burden to a resource. Conducting regular waste audits, setting reduction and recycling targets, and integrating resource recovery into production systems can significantly enhance both environmental sustainability and economic efficiency.

Future-oriented strategies should also explore the role of technology in waste monitoring and control. Integration of IoT-based sensors, AI-driven analytics, and automated sorting systems could enhance real-time tracking and improve decision-making, particularly in large-scale processing units[50].

Another area for exploration is the supply chain dimension, including procurement, storage, packaging, and logistics, which directly influences waste generation

[51]. Aligning waste management practices across the supply chain could strengthen sustainability outcomes. On the consumer side, research into how demand for eco-certified or sustainably processed food products influences company waste policies would add valuable insights.

Equally important is the assessment of social impacts. Poor waste practices often affect the health, income, and quality of life of nearby communities, particularly marginalized groups. Studies examining gender dynamics in waste-handling jobs and initiatives that encourage women's participation in recycling and resource recovery enterprises can help ensure socially inclusive policy design [52].

Finally, comparative studies across Gujarat's regions (such as Saurashtra, Kutch, and South Gujarat) may reveal important variations in waste types and management methods. Quantitative analyses of technologies like pyrolysis, biogas plants, or vermicomposting could further highlight their environmental and economic benefits [53].

By combining government incentives, technological innovation, training, collaboration, and social inclusiveness, Gujarat's Processing of Fruits and Vegetables industry can evolve toward a resilient, efficient, and sustainable waste management system [54].

8 Conclusion

The study on waste disposal and management practices in Fruit And Vegetable Processing Plants in Gujarat provides a detailed understanding of the types and quantities of waste generated, current disposal methods, and the challenges associated with managing such waste effectively. Analysis of the surveyed plants indicates that the majority of waste is organic in nature, predominantly consisting of peels, pulp, seeds, and other biodegradable matter. This conclusion is drawn from direct observations and data collected from the processing units during peak and off-peak production periods. The volume of waste produced varies with the scale of operations, type of raw produce processed, and the efficiency of the production processes [40].

Regarding the assessment of existing waste management practices, the study found a mixed scenario. Larger, well-established processing units have implemented semi-structured waste segregation systems and composting mechanisms, whereas a significant proportion of small and medium-sized enterprises (SMEs) still rely on rudimentary disposal methods such as open dumping, burning, or discharging waste into nearby landfills or water bodies without proper treatment. This finding is based on surveys conducted across 100 processing plants, of which 80 were classified as SMEs, indicating the prevalence of informal waste management practices in smaller units [55].

The study further examined the composition of these processing plants and the types of products they manufacture. Plants vary from fruit pulp and juice units to canned vegetables and frozen produce operations, each generating specific waste streams. For instance, juice extraction units primarily produce peels and pulp, while vegetable processing plants generate trimmings and seeds.

Evaluation of the effectiveness and sustainability of current practices reveals growing awareness among processing units about environmentally sound waste management. Some plants have adopted composting, bio-methanation, or use of by-products for animal feed, resulting in measurable environmental benefits. However, the adoption of these practices is inconsistent, largely due to financial limitations, lack of technical knowledge, and logistical challenges, particularly among smaller units [56].

The study also identifies key barriers hindering efficient waste management, including the high cost of treatment technologies, insufficient trained personnel, limited awareness of government policies and incentives, and lack of space for on-site waste treatment facilities. Moreover, the absence of integration between production operations and waste management systems makes it challenging to implement circular economy approaches within the sector. Resistance to change among small units, often operating on thin profit margins, further complicates efforts toward sustainable practices (Rathod et al., 57).

In conclusion, the study emphasizes the urgent need for a coordinated approach to improve waste management in Gujarat's Processing of Fruits and Vegetables industry. Key recommendations include:

- Strengthening policy-level interventions and regulatory enforcement to promote standardized waste management practices.
- Conducting capacity-building programs and training for plant managers and workers on sustainable disposal techniques.
- Encouraging adoption of low-cost, scalable, and eco-friendly technologies such as composting, bio-methanation, and value-added waste utilization.
- Facilitating knowledge sharing and collaboration between large and small enterprises to enable practical implementation of circular economy models.

By addressing these areas, the sector can significantly reduce environmental impact, enhance worker and community safety, and move toward more sustainable production practices that align with both state and national environmental objectives.

From a practical standpoint, the findings highlight actionable insights for policymakers, processing industries, and environmental managers. The study not only identifies the prevailing gaps in infrastructure and adoption but also offers evidence-based recommendations for advancing circular economy models in Gujarat's agro-processing sector. Thus, the research contributes original knowledge by bridging academic inquiry with applied solutions, offering both theoretical enrichment and practical relevance.

Appendix

Questionnaire on waste generation and management in fruit and vegetable processing plants (Gujarat)

Instructions:

- Please answer all questions honestly.
- Tick (✓) the appropriate option or fill in the blanks where required.
- All information will be kept confidential and used solely for research purposes.

Section A: General Information

1. Name of the plant (optional): _____
2. Location of the plant: _____
3. Type of plant:
 - ☐ Fruit processing
 - ☐ Vegetable processing
 - ☐ Both
4. Year of establishment: _____
5. Number of employees:
 - ☐ <50
 - ☐ 51–100
 - ☐ 101–200
 - ☐ >200
6. Average daily production capacity (in tons): _____

Section B: Types and Quantities of Waste Generated

7. What types of waste are generated in your plant? (Tick all that apply)
 - ☐ Peels/skins
 - ☐ Pulp
 - ☐ Seeds
 - ☐ Stems and leaves
 - ☐ Packaging waste
 - ☐ Others (please specify): _____
8. Approximate quantity of waste generated per day (in kg or tons):
 - ☐ <100 kg
 - ☐ 100–500 kg
 - ☐ 501–1000 kg
 - ☐ >1000 kg
9. Which stage produces the highest amount of waste?
 - ☐ Washing
 - ☐ Peeling/Trimming
 - ☐ Processing/Cutting
 - ☐ Packaging
 - ☐ Storage/Transportation

Section C: Existing Waste Disposal and Management Practices

10. How is the waste from your plant currently managed? (Tick all that apply)
 - ☐ Composting
 - ☐ Vermicomposting
 - ☐ Animal feed
 - ☐ Biofuel production
 - ☐ Landfill
 - ☐ Incineration
 - ☐ Others (please specify): _____
11. Does your plant have a formal waste management policy or plan?
 - ☐ Yes
 - ☐ No
12. Who is responsible for managing waste in your plant?
 - ☐ Dedicated waste management team
 - ☐ Plant manager
 - ☐ Workers/staff
 - ☐ External service provider

13. How often is waste collected and disposed of?

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Other (please specify): _____

Section D: Effectiveness and Sustainability of Current Practices

14. How effective do you think the current waste management methods are?

- ☐ Very effective
- ☐ Effective
- ☐ Neutral
- ☐ Ineffective
- ☐ Very ineffective

15. Are the current waste management practices environmentally sustainable?

- ☐ Yes, fully sustainable
- ☐ Partially sustainable
- ☐ Not sustainable
- ☐ Not sure

16. Have your waste management practices resulted in cost savings or additional revenue (e.g., from compost, biofuel, animal feed)?

- ☐ Yes, significant savings
- ☐ Yes, moderate savings
- ☐ No savings
- ☐ Not applicable

17. Rate the following aspects of your waste management practices (1 = Poor, 5 = Excellent):

Aspect	1	2	3	4	5
Efficiency of waste collection					
Environmental impact					
Cost-effectiveness					
Staff training and awareness					

Section E: Challenges and Barriers

18. What challenges does your plant face in implementing effective waste management? (Tick all that apply)

- ☐ Lack of technical knowledge
- ☐ High cost of equipment
- ☐ Limited space for waste treatment
- ☐ Regulatory constraints
- ☐ Low awareness among employees
- ☐ Difficulty in market for recycled products
- ☐ Others (please specify): _____

19. How significant are these challenges on a scale of 1–5? (1 = Not significant, 5 = Very significant)

Challenge	1	2	3	4	5
Technical knowledge					
Cost of equipment					
Space constraints					
Regulatory compliance					
Employee awareness					
Market for recycled products					

20. Suggestions to improve waste management in your plant:

Constructs and references of the questionnaire

Constructs	References (APA 7th edition)
Awareness & knowledge of waste management	Aslam, Javaid, Munir, & Sultan [58]; Joshi [11], Yoda, Chirawurah, & Adongo [18], Dzekashu & Akoachere [47]
Types and sources of waste in fruit & vegetable processing	Gupta, Trivedi, & Soni [4], Sivaperumal et al. [40], Bhattacharjya, Sahu, Manna, & Patra [7], Tiwari, Singh, Patel, Tiwari, & Kumar [6]
Current waste disposal practices	Narayana [13], Giusti [14], Meidiana & Gamse [20], Remigios [19], Hossain, Ahmed, & Rahman [17]
Barriers and challenges in waste management	Debrah, Teye, & Dinis [39], Sodhi, Singh, & Singh [37]; Pumpinyo & Nitivat-tananon [53], Mukwevho, Radzuma, & Roos [41], Williams, Narra, Antwi, Quaye, & Hagan [44]
Value addition & utilization of waste	Raj, Prasad, & Bansal [28], Dobariya, Mankad, Ramavat, & Joshi [26], Chirumamilla & Singh [8], Sagar, Pareek, Sharma, Yahia, & Lobo [31], Srivastava & Balakrishnan [48]
Circular economy & sustainability approaches	Cárcamo & Peñabaena-Niebles [12], Durdyev, Koc, Tleuken, & Budayan [35]; Zhumadilova & Zhigitova [46], Smol, Szoldrowska, & Duda [34]; Sasso & Filho [56]
Community participation & institutional support	Ampofo [25], Aming-Hayudini, Lartey, & Boateng [22], Adeniyi [24], Unnikrishnan & Singh [43], Gandhi & Namboodiri [29]
Technological interventions & innovations	Geetha & Rajalakshmi [15], Hantoko, Ardhyana, & Wibowo [27], Joshi [2], Sehwat, Khan, Goyal, & Paul [55]; Parekh, Vaidh, Patel, Banerjee, et al. [10]
Policy, regulations & institutional barriers	Edujee, Dyke, & Cains [45], Kerdlap, Low, & Ramakrishna [42]; Nwaogbe, Urhoghide, & Ekpenyong [50]; Emon, Khan, Rahman, & Others [59]
Traditional & emerging methods	Rathod, Sarkar, & Chaudhari [57]; Thakur, Joshi, & Slathia [3]; Raj, Senapati, & Patel [52]

Supplementary Information

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Additional file 1

Author contributions

The author, Varudhini Chirumamilla, has independently undertaken the entire study, encompassing the review of literature, data collection, data analysis, as well as writing and editing the manuscript, which reflects the originality and novelty of the work.

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Data availability

All data generated or analysed during this study are included in this published article [and its supplementary information files].

Declarations

Ethics approval and consent to participate

All experiments were performed in accordance with relevant guidelines and regulations and the study adheres to the Declaration of Helsinki. This research has been approved by the Science Tech Institute, Lucknow, India, ethics committee and all procedures have been carried out in accordance with the guidelines and regulations of Science Tech Institute, Lucknow, India. A formal ethical approval document has been issued by Science Tech Institute Lucknow and is available upon request. Informed consent was obtained from all participants. No respondent was below the age of 18 years. Informed consent was obtained from all individual participants included in the study. None of the participants were below the age of 16.

Consent for publication

The Author hereby provide consent for the publication of the manuscript detailed above, including data contained within the manuscript. All participants provided consent for anonymised data to be used in scientific publications and no minor were involved in our study.

Competing interests

The authors declare no competing interests.

Clinical trial

Not applicable

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